

In this lecture we begin by recalling some fragments of set theory (for more details read appendix A of your course notes and try exercises therein) and move on to systems of linear equations.

Some set theoretic notions

In naïve set theory one defines a set as follows:

Definition 8. *A set is a collection of objects.*

Two sets are equal if and only if they have the same collection of objects. These objects (if any) are called elements of that set. The notion of elements suggests a membership relation “ $\in$ ” and we write $x \in X$ for x is an element of X and $x \notin X$ for x is not an element of X . The set with no elements is called the empty set and is denoted by $\emptyset$.

Examples The collection with two elements $\{0, 1\}$ is a set. The collection of integers, rational numbers, real numbers, complex numbers, and $m \times n$ matrices over real or complex numbers are all examples of sets usually denoted by $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}, M_{m \times n}(\mathbb{R}), M_{m \times n}(\mathbb{C})$ respectively.

Given the membership relation we can define the following:

Definition 9. *A set S is called a subset of set X if for all $s \in S$ we also have $s \in X$ and we write $S \subseteq X$.*

Examples The empty set $\emptyset$ is a subset of any set. The collections of all (real) symmetric, skew-symmetric, diagonal, upper and lower triangular matrices defined in your tutorial 01 are all subsets of $M_{m \times n}(\mathbb{R})$, and $M_{m \times n}(\mathbb{R}) \subseteq M_{m \times n}(\mathbb{C})$, that is, every real $m \times n$ matrix A is also a complex $m \times n$ matrix..

Given a set X and property ϕ , we can construct from it a subset $\{x \in X \mid \phi(x)\}$, that is, the collection of those elements in X that satisfy property ϕ . Symmetric and skew-symmetric matrices satisfy properties $A^T = A$ and $A^T = -A$, respectively and thus could be described as sets $SYM_n(\mathbb{C}) = \{A \in M_{n \times n}(\mathbb{C}) \mid A^T = A\}$ and $SKEW_n(\mathbb{C}) = \{A \in M_{n \times n}(\mathbb{C}) \mid A^T = -A\}$.

Some basic operations on sets

Definition 10. *Given a set X , let S and T be subsets of X .*

We define the union of S and T by $S \cup T = \{x \in X \mid x \in S \text{ or } x \in T\}$

Examples Take $X = \mathbb{Z}$, we have $ODD \cup EVEN = \mathbb{Z}$ where ODD and $EVEN$ denotes collection of odd and even integers, respectively. $\{0, 1\} \cup \{1, 4\} = \{0, 1, 4\}$.

Definition 11. *Given a set X , let S and T be subsets of X .*

We define the intersection of S and T by $S \cap T = \{x \in X \mid x \in S \text{ and } x \in T\}$

Example Take $X = M_{n \times n}(\mathbb{C})$, we have $SYM_n(\mathbb{C}) \cap SKEW_n(\mathbb{C}) = \{\mathbf{0}\}$. An $n \times n$ matrix A which is both symmetric and skew-symmetric should satisfy both equations $A = A^T = -A$, from which we get $2A = \mathbf{0}$ and so A must be the zero $n \times n$ matrix.

Definition 12. Given a set X , let S and T be subsets of X .

We define the set difference of S and T by $S \setminus T = \{x \in X \mid x \in S \text{ and } x \notin T\}$

Example Take $X = S = \mathbb{R}$ and $T = \mathbb{Q}$, then $\mathbb{R} \setminus \mathbb{Q}$ = set of all irrational numbers.

Definition 13. Given two sets X and Y , we define the Cartesian product of X and Y by $X \times Y = \{(x, y) \mid x \in X \text{ and } y \in Y\}$

Examples Taking $X = Y = \mathbb{R}$ we get $\mathbb{R} \times \mathbb{R} = \mathbb{R}^2$ the usual set of ordered pairs of real numbers, $\{0, 1\} \times \{a, b, c\} = \{(0, a), (0, b), (0, c), (1, a), (1, b), (1, c)\}$. Note that the order in which we take the product is important.

Linear equations and systems of linear equations

Definition 14. Let n be a positive integer. A linear equation in variables $x_1, x_2, \dots, x_n$ is an equation that can be written in form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b$$

where $a_1, a_2, \dots, a_n$ and b are constants.

Examples Equations $4x + y = 100$ and $x - 7y + (10 + i)z = 5$ are linear in variables x, y and x, y, z respectively, while equations $2x^2 + 5y = 3$, $\sin(x) + z = 8$, $x + y + \frac{1}{z} = 0$ and $3xy + z = 0$ are not linear.

Definition 15. A solution of the linear equation $a_1x_1 + a_2x_2 + \dots + a_nx_n = b$ is a sequence of numbers $t_1, t_2, \dots, t_n$ such that $a_1t_1 + a_2t_2 + \dots + a_nt_n = b$, that is, if we make substitution $x_1 = t_1, x_2 = t_2, \dots, x_n = t_n$ the equation holds. To solve an equation means to find all its solutions and we shall call the set of the solutions the solution set.

Examples

- The linear equation $7x = \pi$ has the unique solution $x = \frac{\pi}{7}$ and thus its solution set is $\{\frac{\pi}{7}\}$.
- The linear equation $4x - 5y = 20$ can also be written as $y = \frac{4}{5}x - 4$ (or you could also choose to express x in terms of y and proceed similarly, the answers may look different in form but they will describe the same set of solutions). In our case, we thus see that if we substitute x with any real number λ , we get a unique value for y such that the equation holds. So, let λ be any real number, then all solutions of linear equation $4x - 5y = 20$ are of form $x = \lambda, y = \frac{4}{5}\lambda - 4$, this can also be written as $\begin{pmatrix} x \\ y \end{pmatrix} = \lambda \begin{pmatrix} 1 \\ 4/5 \end{pmatrix} + \begin{pmatrix} 0 \\ -4 \end{pmatrix}$ (where $\lambda \in \mathbb{R}$) in vector form and so the solution set is:

$$\{(x, y) \mid \lambda \in \mathbb{R}, x = \lambda \text{ and } y = \frac{4}{5}\lambda - 4\} \text{ or } \left\{ \lambda \begin{pmatrix} 1 \\ 4/5 \end{pmatrix} + \begin{pmatrix} 0 \\ -4 \end{pmatrix} \mid \lambda \in \mathbb{R} \right\}$$

- The linear equation $x - 5y + 8z = 1$ can also be written as $x = 1 + 5y - 8z$ (or you could also choose to express any variable of your choice in terms of the other two variables and proceed similarly). In our case, we see that if we substitute y and z with any real numbers λ and μ , respectively, we get a unique value for x such that the equation holds. So, let λ and μ be any real numbers, then all solutions of linear equation $x - 5y + 8z = 1$ are of form $y = \lambda, z = \mu, x = 1 + 5\lambda - 8\mu$,

this can also be written as $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -8 \\ 0 \\ 1 \end{pmatrix}$ (where $\lambda, \mu \in \mathbb{R}$) in vector form and so the solution set is :

$$\{(x, y, z) \mid \lambda, \mu \in \mathbb{R}, x = 1 + 5\lambda - 8\mu, y = \lambda \text{ and } z = \mu\}$$

$$\text{or } \left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \lambda \begin{pmatrix} 5 \\ 1 \\ 0 \end{pmatrix} + \mu \begin{pmatrix} -8 \\ 0 \\ 1 \end{pmatrix} \mid \lambda, \mu \in \mathbb{R} \right\}$$

Definition 16. A finite list of linear equations in variables $x_1, x_2, \dots, x_n$ is called a system of linear equations or a linear system. A solution of a linear system is a sequence of numbers $t_1, t_2, \dots, t_n$ that is a solution for each of the equation in the system, that is, it satisfies all equations in the system.

Example We give a simpler system of linear equations that is in a triangular form and easy to solve using back-substitution. (In the next lectures we shall study a method of converting any linear system into an equivalent linear system that is simpler to solve).

$$2x_1 + 3x_2 + 4x_3 = 1$$

$$x_2 - 10x_3 = 3$$

$$12x_3 = 60$$

To solve this linear system, we begin by the last simpler equation and proceed by back-substitution, that is, $12x_3 = 60$ implies $x_3 = 5$, and so using the second equation we get $x_2 = 3 + 10x_3 = 3 + 10(5) = 53$ and finally $x_1 = \frac{1}{2}(1 - 3x_2 - 4x_3) = \frac{1}{2}(1 - 3(53) - 4(5)) = -89$ and thus the solution set is $\{(-89, 53, 5)\}$.

Note that a system of linear equations may have no solutions, that is, its solution set is $\emptyset$. Indeed, a linear system with two equations $x + y = 1, x + y = 10$ has no solutions since there exists no pairs of numbers satisfying those two equations simultaneously.

We have seen examples of systems of linear equations with no solutions, a unique solution and infinitely many solutions (linear equations we saw above are in particular a linear system with 1 equation). But are there other possibilities? We shall finish the lecture by showing that these are the only three possibilities, we then call a linear system with no solutions inconsistent and a linear system with infinitely many solutions indeterminate.

Given a linear system with m equations:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n &= b_1 \\ a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n &= b_2 \\ &\vdots \\ a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n &= b_m \end{aligned}$$

We could write it as one matrix equation $A\mathbf{x} = \mathbf{b}$ with:

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix}, \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_m \end{pmatrix} \text{ and } \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix}$$

Computing matrix multiplication $A\mathbf{x}$ and using our definition of equality of two matrices notice that $A\mathbf{x} = \mathbf{b}$ gives us back our linear system. A and $\mathbf{b}$ in equation $A\mathbf{x} = \mathbf{b}$ are called the coefficient matrix and vector of constants respectively. A system of linear equations in which the vector of constants is the zero vector, that is $A\mathbf{x} = \mathbf{0}$ is called a homogeneous linear system. Note that such a linear system has at least one solution, that is, the trivial solution $\mathbf{x} = \mathbf{0}$.

Proposition 6. *Suppose that a linear system with a coefficient A and vector of constants $\mathbf{b}$ has two distinct solutions $\mathbf{t}$ and $\mathbf{s}$, that is $A\mathbf{t} = \mathbf{b}$ and $A\mathbf{s} = \mathbf{b}$ with $\mathbf{t} \neq \mathbf{s}$, then for each $\lambda \in \mathbb{R}$, we have that $\lambda\mathbf{t} + (1 - \lambda)\mathbf{s}$ is also a solution.*

Proof. $A(\lambda\mathbf{t} + (1 - \lambda)\mathbf{s}) = \lambda A\mathbf{t} + (1 - \lambda)A\mathbf{s} = \lambda\mathbf{b} + (1 - \lambda)\mathbf{b} = \lambda\mathbf{b} + \mathbf{b} - \lambda\mathbf{b} = \mathbf{b}$

□

Thus if our linear system has at least two distinct solutions, it will have infinitely many solutions.